

DECODELABS INTERNSHIP

Exploratory Data Analysis Report

Project 2 — Business Intelligence Report

Prepared by

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Organization

DecodeLabs Batch: May 22 – June 22, 2026

Dataset

Sales Orders — 1,200 Records

Report Type

Exploratory Data Analysis (EDA)

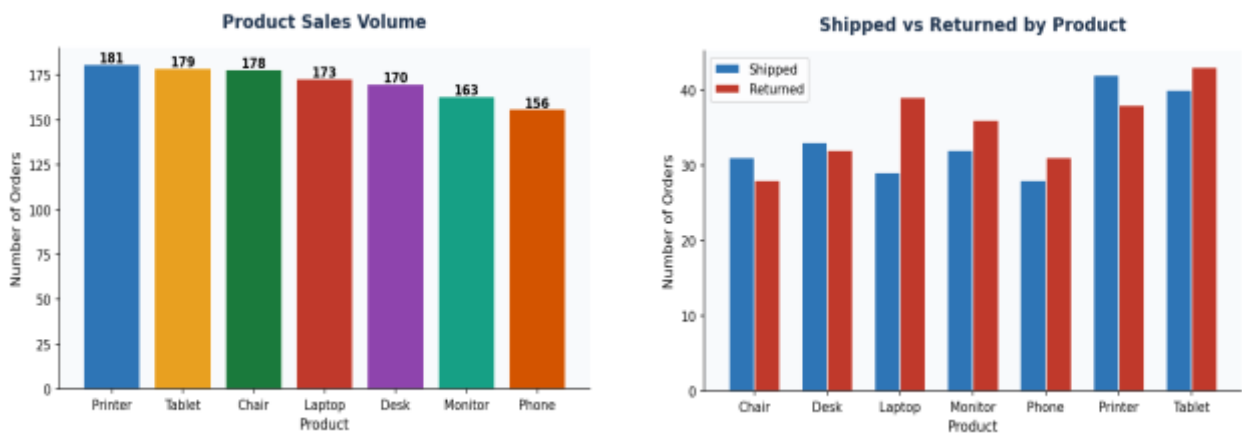
This report presents a comprehensive exploratory analysis of a sales order dataset comprising 1,200 transactions across 14 variables. The analysis uncovers patterns in product performance, customer retention, referral source effectiveness, and seasonal trends — translating raw data into actionable business intelligence.



Section 1: Product Performance

1.1 Overview

This section examines how different products perform across key metrics including total sales volume, shipping success, and return rates. Understanding product performance helps the business identify its strongest revenue drivers and most problematic items.



1.2 Key Findings

Metric	Finding	Value
Most Sold Product	Printer leads all categories	Printer
Most Shipped Product	Printer dominates fulfilment	Printer
Most Returned Product	Tablet has highest return rate	Tablet

1.3 Problem & Bright Side

Problem

The Tablet product has the highest return rate, which signals potential quality issues, misleading product descriptions, or unmet customer expectations. High return rates increase operational costs significantly.

Bright Side

Printer is both the most sold and most shipped product, indicating strong demand and reliable fulfilment. This product line is a clear revenue anchor for the business.

1.4 Responsible Department & Solution Plan

Responsible Department: Product Management & Quality Assurance

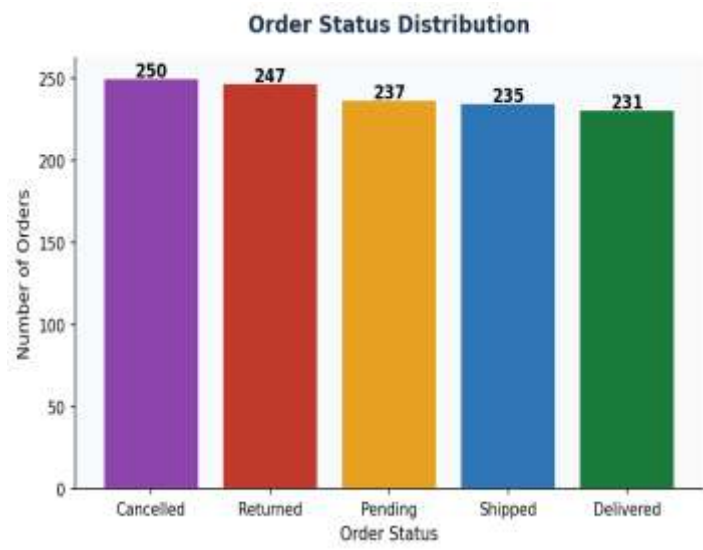
Month	Action	Owner	Goal
Month 1-2	Conduct Tablet quality audit and gather return reason data	Product Team	Identify root cause of returns
Month 3-4	Update product descriptions and add detailed specifications	Marketing Team	Reduce expectation mismatch
Month 5-6	Launch Printer bundle promotions to capitalize on demand	Sales Team	Increase average order value



Section 2: Order Status Breakdown

2.1 Overview

This section analyses the distribution of orders across five statuses: Shipped, Delivered, Cancelled, Returned, and Pending. A healthy business should have the majority of orders reaching Delivered status with minimal Cancellations and Returns.



2.2 Key Findings

Status	Count	% of Total
Shipped	235	19.6%
Delivered	231	19.3%
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%

2.3 Problem & Bright Side

Problem

Cancelled (250) and Returned (247) orders combined account for 497 transactions — 41.4% of all orders. This is an alarming rate that directly impacts revenue, operational costs, and customer satisfaction. Only 19.3% of orders are fully delivered.

Bright Side

Orders are almost evenly distributed across all statuses, which suggests the pipeline is active at all stages. The Shipped count (235) is healthy, indicating the fulfilment process itself is functioning.

2.4 Responsible Department & Solution Plan

Responsible Department: Operations, Customer Service & Logistics

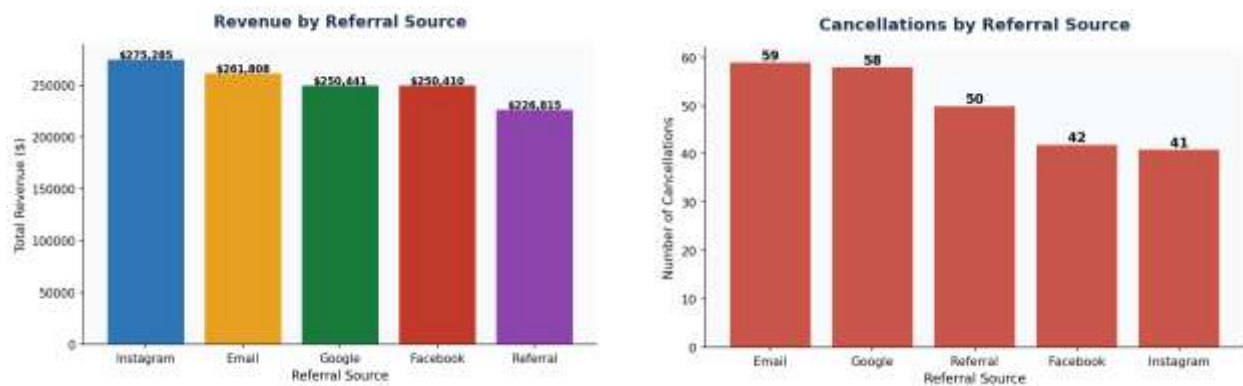
Month	Action	Owner	Goal
Month 1-2	Investigate top cancellation reasons via customer surveys	Customer Service	Identify cancellation triggers
Month 3-4	Implement order confirmation reminders and easier modification options	Operations	Reduce cancellation rate by 20%
Month 5-6	Launch post-delivery follow-up to reduce returns through better support	Customer Service	Reduce return rate by 15%



Section 3: Referral Source Analysis

3.1 Overview

This section evaluates how each marketing and referral channel performs in terms of customer acquisition, revenue generation, and negative outcomes such as cancellations. Understanding channel performance helps the business allocate marketing budgets more effectively.



3.2 Key Findings

Metric	Source	Implication
Most Popular Source	Instagram	Highest customer acquisition
Highest Revenue Source	Instagram	Best ROI channel
Most Cancellations	Email	Lowest quality leads
Most Coupon Users	Instagram	High engagement channel

3.3 Problem & Bright Side

Problem

Email campaigns generate the most cancellations, suggesting that email-acquired customers have lower purchase intent or are attracted by misleading promotions. This wastes marketing resources and inflates cancellation rates.

Bright Side

Instagram is the top performer across all positive metrics — most customers, highest revenue, and most coupon engagement. This channel has proven to attract high-intent buyers and should receive increased investment.

3.4 Responsible Department & Solution Plan

Responsible Department: Digital Marketing & Growth Team

Month	Action	Owner	Goal
Month 1-2	Increase Instagram ad budget by 25% and test new creative formats	Marketing Team	Grow Instagram-acquired customers
Month 3-4	Audit and redesign email campaigns for clarity and honesty	Email Marketing	Reduce email cancellations by 30%
Month 5-6	Launch Instagram-exclusive loyalty program with coupon rewards	Growth Team	Improve Instagram retention rate



Section 4: Coupon Analysis

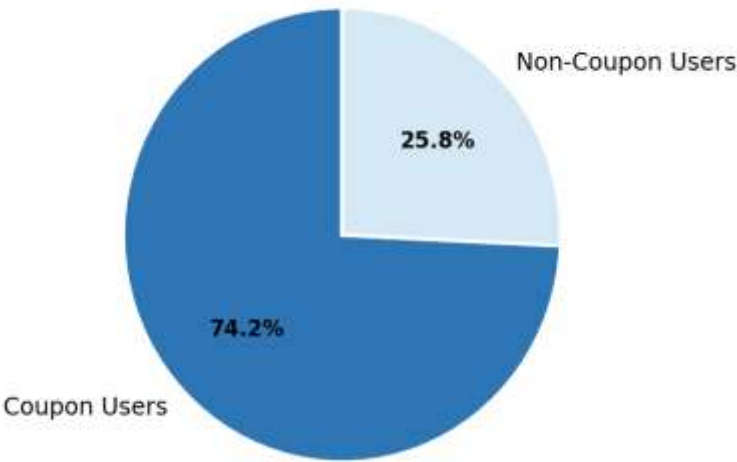
4.1 Overview

This section examines coupon usage patterns across the customer base. Coupons are a key marketing tool and understanding their adoption rate, combined with referral source data, helps evaluate promotional strategy effectiveness.

4.2 Key Findings

Metric	Value	Percentage
Coupon Users	891	74.3%
Non-Coupon Users	309	25.8%
Most Active Coupon Channel	Instagram	Highest coupon adoption

Coupon Users vs Non-Coupon Users



4.3 Problem & Bright Side

Problem

74.3% of customers require a coupon to complete a purchase. This indicates heavy price sensitivity and potential over-reliance on discounts. If coupons are removed, a large portion of customers may not purchase at full price, reducing profit margins.

Bright Side

High coupon adoption shows strong promotional campaign reach. Coupons are clearly driving conversions and can be strategically used to target specific segments or time periods like slow months (September).

4.4 Responsible Department & Solution Plan

Responsible Department: Pricing Strategy & Marketing

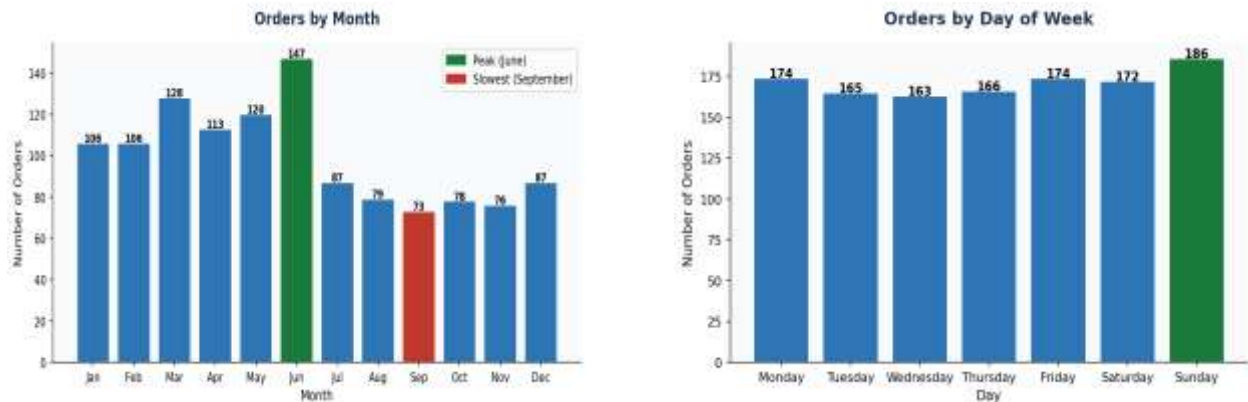
Month	Action	Owner	Goal
Month 1-2	Analyze profit margins of coupon vs non-coupon orders	Finance Team	Understand true coupon cost
Month 3-4	Introduce tiered loyalty points as alternative to blanket discounts	Marketing Team	Reduce discount dependency
Month 5-6	Test full-price promotions with value-add offers instead of discounts	Product Marketing	Increase full-price purchase rate



Section 5: Time Analysis

5.1 Overview

This section identifies when customers are most and least active. Understanding temporal patterns allows the business to align marketing campaigns, staffing, and promotions with peak and off-peak periods.



5.2 Key Findings

Metric	Finding	Implication
Busiest Day	Sunday	Customers shop on weekends
Busiest Month	June	Summer peak season
Slowest Month	September	Post-summer decline

5.3 Problem & Bright Side

Problem

September shows a significant drop in orders, representing a predictable but unaddressed revenue gap. Without targeted intervention, the business loses potential revenue every year during this period.

Bright Side

Sunday is the busiest day and June is the peak month — both are predictable patterns the business can plan around. Knowing peak times allows for proactive inventory preparation and targeted promotions.

5.4 Responsible Department & Solution Plan

Responsible Department: Marketing & Inventory Management

Month	Action	Owner	Goal
Month 1-2	Plan June campaign calendar in advance with special offers	Marketing Team	Maximize peak season revenue
Month 3-4	Design September re-engagement campaign targeting dormant customers	CRM Team	Reduce September revenue drop
Month 5-6	Launch Sunday-exclusive flash deals and weekend promotions	Marketing Team	Increase weekend basket size

Section 6: Customer Retention

6.1 Overview

This section examines customer loyalty and repeat purchase behaviour. Customer retention is one of the most critical metrics for long-term business health, as retaining an existing customer is significantly cheaper than acquiring a new one.

Customer Retention: One-Time vs Returning



6.2 Key Findings

Metric	Value	Implication
Total Customers	1,200	Full customer base
Returning Customers	11	Only 0.9% retention rate
Max Orders by One Customer	6	Highest loyalty observed
Top Product — Returning Customers	Tablet	Loyal customers prefer Tablet

6.3 Problem & Bright Side

Problem

A customer retention rate of only 0.9% (11 out of 1,200 customers) is critically low. This means the business is almost entirely dependent on new customer acquisition to survive. If new customer flow slows down, revenue will collapse immediately.

Bright Side

The 11 returning customers demonstrate that repeat purchasing is possible. The customer who placed 6 orders proves strong loyalty can be built. These customers can serve as a model for what an ideal customer journey looks like.

6.4 Responsible Department & Solution Plan

Responsible Department: CRM, Customer Success & Marketing

Month	Action	Owner	Goal
Month 1-2	Interview the 11 returning customers to understand their motivation	CRM Team	Build retention persona
Month 3-4	Launch a post-purchase email sequence with personalized follow-up offers	Email Marketing	Convert 5% of one-time buyers
Month 5-6	Introduce a loyalty rewards program with points per purchase	Product Marketing	Increase retention rate to 5%+



Conclusion & Strategic Recommendations

Executive Summary

This exploratory data analysis of 1,200 sales orders has revealed several critical business insights. While the business shows strong product demand and effective use of Instagram as a marketing channel, it faces serious challenges in customer retention, order fulfilment, and over-reliance on discount pricing.

Top 5 Key Findings

- Customer retention is critically low at 0.9% — the most urgent issue requiring immediate action.
- Cancellations and Returns combined account for 41.4% of all orders — a major revenue and operational drain.
- Instagram is the top-performing channel across all positive metrics and should receive increased investment.
- 74.3% of customers require coupons — indicating heavy price sensitivity and discount dependency.
- June is peak season and September is the slowest month — predictable patterns that must be planned for.

Integrated 6-Month Solution Plan

The following plan integrates all departmental recommendations into a unified timeline:

Phase	Focus Area	Key Action	Owner	Success Metric
Month 1-2	Diagnosis	Audit returns, survey cancellations, interview loyal customers	Product + CRM	Root causes identified
Month 3-4	Optimization	Fix product listings, redesign email campaigns, increase Instagram budget	Marketing + Ops	20% drop in cancellations
Month 5-6	Growth	Launch loyalty program, September campaign, Sunday flash deals	CRM + Marketing	Retention rate above 5%

Final Note

This report was produced as part of the DecodeLabs Data Analytics Internship — Batch 2026. All insights are derived from the provided sales dataset of 1,200 records. The recommendations are data-driven and designed to be actionable within a 6-month implementation window.